

Xinbao Qiao

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EDUCATION

The Chinese University of Hong Kong
PhD student in Information Engineering

Hong Kong
2026–present

Zhejiang University
M.Sc. in Artificial Intelligence

Zhejiang
2022–2025

- Major GPA: 90/100; rank: 3/25.
- Selected coursework: AI Algorithms and Systems (100), Secure Artificial Intelligence (97).

Shandong University
B.Eng. in Communication Engineering

Shandong
2018–2022

- Major GPA: 82.47/100.
- Third-class Academic Award, 2018–2019.

RESEARCH INTERESTS

- lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted.
- The intersection of AI with network, including AI for communication, communication for AI.

PUBLICATIONS

- [1] **When Sample Selection Bias Precipitates Model Collapse.**
Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang.
Forty-Third International Conference on Machine Learning, ICML, 2026.
- [2] **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness.**
Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang.
Fortieth AAAI Conference on Artificial Intelligence, AAAI, 2026.
- [3] **Hessian-Free Online Certified Unlearning.**
Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei.
Thirteenth International Conference on Learning Representations, ICLR, 2025.
- [4] **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest.**
Shurong Wang, Zhuoyang Shen, Xinbao Qiao, Tongning Zhang, Meng Zhang.
Thirteenth International Conference on Learning Representations, ICLR, 2025.
- [5] **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification for Pretrained Model.**
Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang.
Under review. Fortieth Conference on Neural Information Processing Systems, NeurIPS, 2026.
- [6] **Illusory Pattern Perception Drives Spurious Inference in Large Language Models.**
Peihua Mai, Zhuoyan Shao, Xinbao Qiao, Meng Zhang, Xinyue Zhou, Yan Pang.
Under review. Fortieth Conference on Neural Information Processing Systems, NeurIPS, 2026.
- [7] **Learn What Matters: Data Pruning for Efficient Decentralized Learning.**
Xinbao Qiao, Xunhao Jiang, Zuozhu Liu, Peng Sun, Meng Zhang.
Under review.

RESEARCH EXPERIENCE

Research on Data-Centric ML Systems

Advisor: Prof. Meng Zhang, Zhejiang University

M.Eng. Student, 03/2023 - 12/2025

- **Data Influence Attribution.** Developed theoretically guaranteed approaches to quantify and analyze the influence of data on learning models, and designed unlearning techniques to remove these influences.
- **Trade-offs in Trustworthy Artificial Intelligence.** Investigated and developed data-centric methodologies to balance fairness, robustness, privacy, and utility in machine learning systems.

- **Cloud-Edge Collaborative Human-Space Healthcare.** Architected cloud-edge collaborative frameworks for real-time multimodal data monitoring of human health using video, audio, and sensor signals.

Research on Trustworthy LLM systems

Advisor: PANG Yan, James, National University of Singapore Full-time Research Intern, 06/2025 - 12/2025

- **Illusory Pattern Perception in LLMs.** Examined whether LLMs exhibit illusory pattern perception, a known human cognitive bias, and offered an interpretable framework to account for LLM behavior.
- **Synthetic Data Evaluation.** Developed a distributed Wasserstein collaborative algorithm, providing a private Wasserstein methodology for synthetic data evaluation when real data is limited.

SKILLS

Programming: Python, PyTorch, TensorFlow, MATLAB, C/C++, SQL, Linux, Docker, Git, $\text{\LaTeX}$.

Languages: Mandarin Chinese (native), English.